

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks: 50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

Solution:

Problem Understanding

The given problem involves estimating the probability of getting **Heads** using the **Maximum Likelihood Estimation (MLE)** method. MLE estimates the parameter that best explains the observed data. Since the experiment follows a Bernoulli process, the MLE is calculated by dividing the number of successful outcomes (Heads) by the total number of observations.

Given

- Total number of coin tosses (n) = 10
- Number of Heads (H) = 7
- Number of Tails (T) = 3

Method Selection

For a Bernoulli experiment, the Maximum Likelihood Estimator of the probability is:

$$\hat{p} = \frac{\text{Number of Successful Outcomes}}{\text{Total Number of Observations}}$$

Calculation

Substitute the given values into the formula.

$$\hat{p} = \frac{7}{10}$$
$$\hat{p} = 0.7$$

Answer

The Maximum Likelihood Estimate (MLE) of the probability of getting Heads is:

$$\hat{p} = 0.7$$

or **70%**.

Interpretation of Results

The estimated probability of obtaining Heads is **0.7**. This means that, based on the observed outcomes, the coin produced Heads **70%** of the time. This estimate is obtained from the available sample and represents the most likely value of the probability according to the MLE method.

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Solution:

Problem Understanding

This problem asks us to estimate the probability that a student passes an exam using **Maximum Likelihood Estimation (MLE)**. Passing the exam is considered a success, while failing is considered a failure. Since each student has two possible outcomes, the Bernoulli distribution is applicable.

Given

- Total number of students = **50**
- Number of students passed = **42**
- Number of students failed = **8**

Method Selection

The MLE for the probability of success is:

$$\hat{p} = \frac{\text{Number of Successes}}{\text{Total Number of Observations}}$$

Calculation

$$\begin{aligned}\hat{p} &= \frac{42}{50} \\ \hat{p} &= 0.84\end{aligned}$$

Answer

The Maximum Likelihood Estimate of the probability that a student passes is:

$$\hat{p} = 0.84$$

or **84%**.

Interpretation of Results

The estimated probability of passing the examination is **84%**. This indicates that, based on the sample data, approximately **84 out of every 100 students** are expected to pass the exam.

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.

Find the MLE for the probability that a bulb is defective.

Solution:

Problem Understanding

The objective is to estimate the probability that a light bulb is defective using the Maximum Likelihood Estimation method. Here, a defective bulb is considered a success for the Bernoulli experiment.

Given

- Total number of bulbs = **100**
- Number of defective bulbs = **8**
- Number of non-defective bulbs = **92**

Method Selection

The MLE formula for estimating probability is:

$$\hat{p} = \frac{\text{Number of Defective Bulbs}}{\text{Total Number of Bulbs}}$$

Calculation

$$\hat{p} = \frac{8}{100}$$
$$\hat{p} = 0.08$$

Answer

The Maximum Likelihood Estimate of the probability that a bulb is defective is:

$$\boxed{\hat{p} = 0.08}$$

or **8%**.

Interpretation of Results

The estimated defect rate is **8%**. This means that, according to the observed sample, about **8 bulbs out of every 100 bulbs** are expected to be defective.

4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.

Solution:**Problem Understanding**

The problem requires estimating the probability of obtaining the outcome **6** when rolling a biased die. Since rolling a 6 is treated as a success, the Maximum Likelihood Estimation method can be used.

Given

- Total number of rolls = **60**
- Number of times 6 appeared = **18**
- Number of other outcomes = **42**

Method Selection

The MLE formula is:

$$\hat{p} = \frac{\text{Number of Times 6 Appeared}}{\text{Total Number of Rolls}}$$

Calculation

$$\hat{p} = \frac{18}{60}$$
$$\hat{p} = 0.3$$

Answer

The Maximum Likelihood Estimate of rolling a **6** is:

$$\boxed{\hat{p} = 0.3}$$

or **30%**.

Interpretation of Results

The estimated probability of rolling a 6 is **30%**, which is greater than the probability for a fair die ($1/6 \approx 16.7\%$). Therefore, the sample data suggests that the die is **biased**.

5. Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Solution:

Problem Understanding

The problem involves calculating the predicted output, finding the prediction error, and updating the weight using the gradient descent learning rule. This process is commonly used in machine learning to reduce prediction errors.

Given

- Input (x) = 4
- Weight (w) = 2
- Actual Output = 10
- Gradient = 4
- Learning Rate = 0.01

Step 1: Find the Predicted Output

Formula

$$y = w \times x$$

Calculation

$$y = 2 \times 4 = 8$$

Answer

Predicted Output = 8

Step 2: Calculate the Error

Formula

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Calculation

$$\text{Error} = 10 - 8 = 2$$

Answer

Prediction Error = 2

Step 3: Update the Weight

Formula

$$w_{\text{new}} = w - (\text{Learning Rate} \times \text{Gradient})$$

Calculation

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04$$

$$w_{\text{new}} = 1.96$$

Answer

Updated Weight = **1.96**

Interpretation of Results

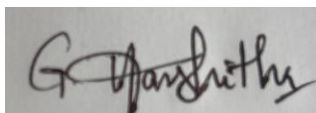
The model initially predicted an output of **8**, while the actual output was **10**, resulting in an error of **2**. By applying gradient descent, the weight is updated from **2.00** to **1.96**. This adjustment helps reduce the prediction error and improves the model's performance in future predictions.

Code of Conduct:

I (G.Harshitha / Reg No:192372384) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation